# Embeddings in Machine Learning

## What Are Embeddings?

Embeddings are dense, low-dimensional vector representations of high-dimensional data such as words, sentences, images, or users. Instead of representing data as sparse one-hot vectors, embeddings map each item to a continuous point in a learned vector space.

The key insight is that semantically similar items end up close together in this space. For example, the words 'king' and 'queen' will have vectors that are geometrically close, reflecting their shared semantic relationship.

## Word Embeddings

Word2Vec, introduced by Mikolov et al. in 2013, was one of the first popular embedding methods. It trains a shallow neural network on a large corpus to predict a word from its neighbors (CBOW) or predict neighbors from a word (Skip-gram).

GloVe (Global Vectors for Word Representation) is another approach that factorizes a word co-occurrence matrix to produce embeddings that capture both local and global statistical information.

FastText extends Word2Vec by representing each word as a bag of character n-grams, making it robust to rare and misspelled words.

## Sentence and Document Embeddings

While word embeddings capture individual word meaning, many tasks require representing entire sentences or documents. Sentence-BERT (SBERT) fine-tunes BERT using siamese networks to produce semantically meaningful sentence embeddings.

Universal Sentence Encoder (USE) from Google provides embeddings optimized for tasks like semantic textual similarity, clustering, and classification.

## How Embeddings Are Trained

Embeddings are typically learned as a by-product of training a neural network on a proxy task. For language, the proxy task might be predicting masked tokens (as in BERT) or next-token prediction (as in GPT).

The embedding layer is simply a lookup table: a matrix of shape (vocabulary_size, embedding_dim). During training, gradients update these vectors so that the model performs better on the task, incidentally encoding semantic structure.

## Similarity and Distance

Once embeddings are trained, similarity between items is measured geometrically. The most common metric is cosine similarity: cos(a, b) = (a . b) / (||a|| * ||b||). Two vectors pointing in the same direction have cosine similarity of 1; orthogonal vectors have similarity 0.

Euclidean distance is also used, particularly when the magnitude of vectors matters. For most NLP retrieval tasks, cosine similarity is preferred because it is scale-invariant.

## Applications

Semantic search: A query is embedded into the same space as documents. The most relevant documents are those whose embeddings are closest to the query embedding.

Retrieval-Augmented Generation (RAG): Embeddings power the retrieval step, fetching relevant context from a knowledge base before passing it to a generative model.

Recommendation systems: Users and items are embedded into a shared space; recommendations are items closest to a user's embedding.

Clustering and classification: Embeddings serve as feature vectors for downstream models, reducing the need for manual feature engineering.

## Vector Databases

At scale, finding the nearest neighbors in a large embedding space requires specialized infrastructure. Vector databases such as Pinecone, Weaviate, Chroma, and FAISS implement Approximate Nearest Neighbor (ANN) algorithms like HNSW and IVF-PQ.

These systems allow millions of embeddings to be indexed and queried in milliseconds, making large-scale semantic search and RAG pipelines practical.